CHILD DEVELOPMENT PERSPECTIVES

Adolescent Personality Development and Identity Formation

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ABSTRACT—***Personality development is not only about changes in traits but also about changes in other layers of the self, such as the identity layer. Forming one's identity is thought to be the key developmental task of adolescence, but profound changes in personality traits also occur in this period. In this article, I summarize research on these processes, including research on adolescent personality trait development from a variable-centered and a typological point of view. With regard to identity formation, I also describe contemporary models to measure identity formation and address the importance of examining identity formation on a day-to-day level. Despite obvious theoretical ties between personality traits and identity formation, few studies have examined how these processes affect one another across time.***

KEYWORDS—*adolescence; personality; identity formation*

Adolescence is a time when the self-system is redefined (Erikson, 1950). The core of this self-system consists of personality traits that emerge in the first years of life (McAdams & Olson, 2010). These traits capture the ways in which individuals' characters differ. From childhood onward, motivations and goals also start to guide behavior. In adolescence, when individuals gain the cognitive capacities to engage in abstract thinking, they begin to search for sameness and continuity of the self. In other words, adolescents are expected to form an identity that provides meaning to life. Identity formation is thought to be the key developmental task of adolescence (Erikson, 1950).

Different aspects of the self emerge in different periods of the lifespan. However, once new aspects of the self emerge, existing aspects do not stop developing. Therefore, it is important to consider several aspects of the self. In this article, I focus on adolescence, and consider the core of the self (i.e., personality traits) and the core developmental task (i.e., identity formation) of this period. Personality traits and identity dimensions are not only of theoretical interest; they are also important because of their strong ties to psychological adjustment (e.g., Caspi, Roberts, & Shiner, 2005; Crocetti, Rubini, & Meeus, 2008). Exciting research is being conducted on both personality development and identity formation, including new approaches to adolescent personality development and new research on identity formation.

CURRENT APPROACHES TO ADOLESCENT PERSONALITY DEVELOPMENT

Changes in Traits

The Big Five traits are thought to capture the core of personality: Neuroticism (i.e., the tendency to experience stress), Extraversion (i.e., the tendency toward positive emotionality and social dominance), Openness to Experience (i.e., curiosity, creativity, and imagination), Agreeableness (i.e., helpfulness, cooperativeness, and kindness), and Conscientiousness (i.e., orderliness, responsibility, and perseverance; Caspi et al., 2005; McCrae & John, 1992). Personality traits have long been regarded as virtually unchangeable and solely determined by genes, but today, traits are commonly perceived as relatively stable constructs that can be affected by environmental factors such as social roles and relations (Roberts, Wood, & Smith, 2005).

Recently, several large-scale longitudinal studies have shed light on adolescent personality development. Acknowledging that development encompasses more than mean-level changes, these studies examined rank-order stability and profile stability.

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Rank-order stability indicates to what extent individual differences in particular traits are stable across time. Profile stability indicates to what extent the relative positioning of traits within particular individuals is maintained across time. For instance, someone could be more conscientious than agreeable and more agreeable than extraverted. This person would reflect high-profile stability if the same order was maintained 1 year later. Thus, profile stability reflects how consistently personality trait profiles (i.e., constellations of multiple traits) are organized.

Adolescence usually involves changes toward an adult-like personality profile, with the expectation that Neuroticism will decrease and Agreeableness, Conscientiousness, Extraversion, and Openness will increase (e.g., Caspi et al., 2005). In addition, individual differences in personality traits become more set (e.g., Costa & McCrae, 1994), and personality trait profiles stabilize (e.g., Roberts, Caspi, & Moffitt, 2001). A longitudinal study of individuals ages 12–20 (Klimstra, Hale, Raaijmakers, Branje, & Meeus, 2009) found clear evidence for such changes, specifically, mean-level increases in Agreeableness, Extraversion, and Openness, and decreases in Neuroticism. Rank-order stability and profile stability also increased systematically. Overall, girls were ahead of boys in these developmental processes.

Other studies (De Fruyt et al., 2006; Van den Akker, Dekovic, & Prinzie, 2010) have found decreases in Conscientiousness and no increases in the other Big Five traits from middle childhood to early adolescence. A large-scale (N = 1,267,218) cross-sectional study (Soto, John, Gosling, & Potter, 2011) reconciled these seemingly conflicting results, demonstrating that changes toward adult-like levels occur only after early adolescence. Still, despite our increased understanding of adolescent personality trait development, to fully unravel the process, we need longitudinal studies covering the entire period of childhood to early adulthood with frequent (i.e., yearly) measurements. Moreover, we know relatively little about antecedents and correlates of individual differences in changes in personality traits. Personality traits are linked to three areas—social relationships, education and occupation, and general health (Caspi et al., 2005). Although we know, for example, that individuals with low levels of Agreeableness are at greater risk for disease, few studies have examined the direction of effects between personality traits and variables reflecting adaptation in the aforementioned domains.

Typological Approaches

Approaches that consider traits individually ignore the fact that two individuals with the same score on one trait can differ depending on their score on another trait. For example, highly agreeable extraverts are considered gregarious, whereas nonagreeable extraverts are thought to be arrogant (Wiggins, 1979). Therefore, personality psychologists should use typological approaches that consider constellations of traits within individuals (e.g., Allport, 1937; Asendorpf, Borkenau, Ostendorf, & van Aken, 2001).

Block (1971) distinguished five personality types; three of them were later largely replicated (Robins, John, Caspi, Moffitt, & Stouthamer-Loeber, 1996): resilients (i.e., generally well-adjusted individuals), undercontrollers (i.e., individuals with a tendency toward externalizing problems), and overcontrollers (i.e., individuals with a tendency toward internalizing problems). The three types displayed distinct Big Five personality profiles: Resilients had low levels of Neuroticism and high levels on the other Big Five traits, undercontrollers had low levels of Agreeableness and Conscientiousness, and overcontrollers were Neurotic and had low levels of Extraversion. These personality types were replicated in subsequent studies (for an overview, see Asendorpf et al., 2001).

The status of personality types is by no means undisputed (e.g., Chapman & Goldberg, 2011). Proponents of trait approaches have shown that resilients, undercontrollers, and overcontrollers are not always replicable and account for much less variance in outcomes than personality traits (e.g., Ashton & Lee, 2009). However, other studies have found that types were replicable in different age groups, with different statistical techniques, with different instruments (e.g., Hart, Atkins, & Fegley, 2003), and regardless of whether broad Big Five traits or the narrower traits (i.e., facets) underlying these broad traits were used as input (Klimstra, Luyckx, Teppers, Goossens, & de Fruyt, 2011). Furthermore, personality types appeared to have incremental validity as they predicted outcomes above and beyond personality traits (Asendorpf & Denissen, 2006). Thus, despite their somewhat controversial status, personality types appear to be replicable and useful constructs.

Resilients, undercontrollers, and overcontrollers are especially popular in research on adolescence. For example, childhood personality types were excellent predictors of adolescent adjustment (i.e., shyness and aggression) and the timing of demographic transitions (i.e., leaving home, establishing a romantic relationship, and getting a job; Denissen, Asendorpf, & van Aken, 2008). Resilients were the best adjusted and the earliest to make the aforementioned transitions. Overcontrollers were relatively shy and undercontrollers were relatively aggressive. Both types transitioned later than resilients and, partly as a result, became more aggressive as they grew older.

Personality types can also be used to study personality development, either by looking at transitions between personality types or examining developmental trajectories of personality. Studies on personality type transitions (Akse, Hale, Engels, Raaijmakers, & Meeus, 2007; Asendorpf et al., 2001; van Aken & Dubas, 2004) have found that about 50% of adolescents changed from one type to another. However, these studies did not examine developmental patterns in the observed transitions. A study that did (Meeus, van de Schoot, Klimstra, & Branje, 2011) found that most individuals (73.5%) did not change from one type to another. However, the transition patterns among individuals that did change type suggest that adolescents changed toward more adult-like personality profiles, with the most common transitions from undercontroller and overcontroller to the resilient type. As a result, the percentage of resilients increased,

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the percentage of overcontrollers dropped slightly, and undercontrollers almost disappeared as a group. Changes in personality type were accompanied by changes in problem behavior symptoms. Overcontrollers who changed into resilients became less anxious, whereas individuals who moved in the opposite direction became more anxious.

Another study (Klimstra, Hale, Raaijmakers, Branje, & Meeus, 2010a) followed Block's (1971) initial recommendations to focus on a developmental typology. Rather than examining transitions from one type to another (which is not possible with this approach), developmental typologies focus on classifying individuals in groups that display distinct developmental trajectories. In this case, resilients, undercontrollers, and overcontrollers were replicable as developmental trajectories (Klimstra et al., 2010a); that is, three types reflected profiles similar to those obtained with cross-sectional methods in terms of mean levels. These types were also among the ones identified in previous studies on developmental trajectories of personality trait profiles (Block, 1971; Morizot & Le Blanc, 2005).

In the Klimstra study (Klimstra et al., 2010a), resilients, undercontrollers, and overcontrollers remained clearly distinguishable from one another across time, so, for example, the relatively nonconscientiousness and nonagreeable undercontrollers became more conscientious and agreeable as they grew older. However, they were still less conscientious and agreeable than resilients because resilients increased about as much in Conscientiousness and Agreeableness as undercontrollers did.

In the Meeus study (Meeus et al., 2011), these changes would have led to different conclusions; that is, undercontrollers who increased in Agreeableness and Conscientiousness past a certain threshold would have joined the resilients. However, individuals who already were resilient and had similar amounts of increases would also remain in this category because additional categories to indicate more mature personality profiles did not exist. As a result, undercontrollers almost disappeared in the Meeus study, which contradicts the notion that resilients, undercontrollers, and overcontrollers are replicable at different ages (Asendorpf et al., 2001; Hart et al., 2003). More importantly, valuable information with regard to individual differences in personality change might get lost. The method used by Klimstra et al. (2010a) also has drawbacks because individuals can never transit from one type to another. Some individuals undergoing personality changes do not adhere to the general developmental patterns (Meeus et al., 2011). If relatively few individuals undergo such changes, or if these changes appear in a limited number of traits, it is hard to identify such individuals as a separate group. Thus, both typological approaches to personality development have drawbacks. A trait approach considering both mean-level change and individual-level change would have dealt with these problems. However, as noted previously, trait approaches focus only on one trait at a time and provide no information on how the overall personality trait profiles of particular individuals change. One way to reconcile the limitations of all of the approaches may be to use Meeus et al.'s (2011) transition approach while correcting for general developmental trends shared by all personality types when constructing these types at different ages. Thus, if 12-year-old resilients had a mean score of 4 on Agreeableness, and analyses indicated that mean levels of this trait had increased with 1 point between ages 12 and 20 across the whole sample (i.e., including resilients, undercontrollers, and overcontrollers), then individuals should be classified as resilients at age 20 only if they exhibit an Agreeableness score of 5 or more. In other words, different norms should be used to classify someone as a resilient at different ages.

LINKS BETWEEN PERSONALITY TRAIT DEVELOPMENT AND IDENTITY FORMATION

Adolescents make important changes toward an adult-like personality. However, adolescence also marks the gradual emergence of another layer of the self: identity (Erikson, 1950; McAdams & Olson, 2010). In fact, identity processes might cause some of the changes in adolescents' personalities. Specifically, increased identification with adult social roles (i.e., commitment), which is a benchmark of identity formation, has been proposed as a driving force behind adolescent personality change in the so-called social investment principle (e.g., Roberts et al., 2005). Indicators, but not direct measures, of identity commitment have been shown to be cross-sectionally associated with the personality traits Conscientiousness, Agreeableness, and Neuroticism (Lodi-Smith & Roberts, 2007), but few studies have examined these associations longitudinally. In addition, the links between personality and identity formation can be fully understood only if we understand identity formation itself.

IDENTITY FORMATION: TOWARD EXAMINING DAY-TO-DAY PROCESSES

Identity is a broad concept (for an overview, see Schwartz, Luyckx, & Vignoles, 2011); in this article, I consider only identity formation. Erikson (1950) described identity formation as a process in which childhood identifications are replaced by, or reinterpreted as, one's own self-defined set of commitments. Marcia (1966) proposed that this process was guided by commitment (i.e., choosing alternatives in important life domains and engaging in activities to implement these choices) and exploration (i.e., comparing alternatives before making a choice). This conceptualization nicely captures the identity processes described in the social investment principle (Roberts et al., 2005).

Recently, Marcia's (1966) model has been extended in several ways, with an emphasis on the multifaceted nature of exploration (Crocetti et al., 2008; Luyckx et al., 2008). Of the two types of exploration processes, a first type involves the comparison of several alternatives such as exploration in breadth (i.e., comparing several alternatives before making choices; Luyckx et al., 2008) and reconsideration (i.e., doubting current commitments

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and comparing them with alternatives; Crocetti et al., 2008). A second type involves reflection on the merits of one's current commitments (Crocetti et al., 2008; Luyckx et al., 2008) such as in-depth exploration.

These extended identity formation models have uncovered several important developmental phenomena, such as intertwined commitment-formation and commitment-evaluation cycles (Luyckx, Goossens, & Soenens, 2006). In the former, identity commitments are established after a period of exploration in breadth. The latter refers to the evaluation of one's commitments, which could possibly lead to a search for new commitments and a move back to the commitment-formation cycle. Moreover, commitment-formation processes, evidenced by decreasing reconsideration, were found to be more prominent in early adolescence (Klimstra, Hale, Raaijmakers, Branje, & Meeus, 2010b), whereas identity-evaluation processes, evidenced by increasing in-depth exploration, gained importance only in late adolescence. These studies relied on longitudinal data with relatively long intervals (i.e., at least 6 months) between measurements, which is limiting because identity formation is a dynamic process rooted in emotions and developing in relationships (Kunnen, Bosma, van Halen, & van der Meulen, 2001). To capture this dynamic process, identity formation should be gauged with much shorter intervals between measurements.

Klimstra, Luyckx, et al. (2010) tracked two key processes of early adolescent identity formation (i.e., commitment and reconsideration) from day to day to find that reconsideration and commitment predicted one another between one day to the next. This so-called commitment-reconsideration dynamic was strikingly similar to Erikson's (1950) theory, which described identity formation as a process in which adolescents move between a sense of identity (i.e., commitment) and role confusion (i.e., reconsideration). Reshaping one's identity most likely starts with reconsideration, which predicts decreases in commitment. Subsequently, this weakened sense of commitment may lead to further reconsideration. Klimstra, Luyckx, et al. also found that the amount of commitment and reconsideration varied across days. Day-to-day fluctuations in reconsideration predicted a weaker identity, with lower levels of commitment and higher levels of reconsideration. This suggests that inconsistently doubting commitments, evidenced by fluctuations in reconsideration, will eventually weaken one's identity. A subsequent study (Schwartz et al., 2011) showed that greater fluctuations in identity dimensions also predicted increases in symptoms of internalizing problem.

Short-term processes themselves are important, but Kunnen et al. (2001) and even Erikson (1950) also pointed to the importance of context by referring to the relevance of relationships in identity development. Studies have shown the effects of relationships on personality traits (e.g., decreases in Neuroticism after transitioning to a romantic relationship; e.g., Neyer & Lehnart, 2007). Research also should examine possible relationship effects on identity formation.

CONCLUSION

Research on the development of personality traits and types and on identity formation in adolescence has evolved in recent years, with changes toward adult-like levels seen in all three areas. In addition, studying day-to-day processes seems crucial to understand identity formation. Still, there is relatively little knowledge on the determinants, correlates, and consequences of personality and identity change, and research on the day-to-day processes involved in identity formation is still in its infancy.

McAdams and Olson (2010) pointed to the importance of examining how different parts of the self-system (i.e., identity dimensions and personality traits) affect one another, but few studies address that topic. Luyckx, Soenens, and Goossens (2006) found that global identity dimensions (i.e., summarizing educational and relational identity) and personality traits mutually affected one another across time. For example, openness was a predictor of and was predicted by exploration in breadth. Klimstra, Luyckx, Germeijs, Meeus, and Goossens (2012) showed that links between personality traits and educational identity dimensions were unidirectional, from personality to identity. Conscientiousness was the best predictor of educational identity formation. Thus, personality traits and identity formation dimensions affect each other, but how they do so seems to depend on the identity domain considered. In addition, previous studies considered only single traits; it remains unclear how personality types are associated with identity dimensions.

Although our understanding of personality traits, personality types, and identity dimensions has grown, more research is needed, especially studies linking different parts of the self-system to infer the exact nature of their associations. It would be especially useful to link personality traits and types to separate identity domains while considering daily measurement of identity dimensions.

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